

Source URL: https://en.wikipedia.org/wiki/Airbus_A380

Query: what is airbus_a380

Answer: # Airbus A380 Documentation

Overview

The Airbus A380 is a large wide-body airliner developed and produced by Airbus until 2021. It is the world's largest passenger airliner and the only full-length double-deck jet airliner ever to enter production.

Key Features:

- * **Double-Deck Configuration:** Full-length double deck for maximized passenger capacity.
- * **Wide-Body Airliner:** Offers spacious seating and cargo options.
- * **High Capacity:** Typical seating for 525 passengers, with a maximum certified capacity for 853 passengers.
- * **Long Range:** Range of 8,000 nmi (14,800 km; 9,200 mi).
- * **Engines:** Powered by either Engine Alliance GP7200 or Rolls-Royce Trent 900 turbofans.

Image

![Emirates

A380](https://upload.wikimedia.org/wikipedia/commons/thumb/0/09/A6-EDY_A380_Emirates_31_jan_2013_jfk_%288442269364%29_%28cropped%29.jpg/330px-A6-EDY_A380_Emirates_31_jan_2013_jfk_%288442269364%29_%28cropped%29.jpg)

Image description: An Emirates A380 aircraft is shown in flight.

Development History

Origins:

- * **1988:** Airbus began studies to challenge the dominance of the Boeing 747 in the long-haul market.
- * **1990:** Project announced at the Farnborough Airshow.
- * **1994:** A3XX project presented.
- * **December 19, 2000:** The Airbus supervisory board launched the €9.5 billion A380 program.

Production:

- * **January 18, 2005:** First prototype unveiled in Toulouse, France.
- * **April 27, 2005:** First flight.
- * **December 12, 2006:** Type certificate obtained from EASA and FAA.
- * **October 15, 2007:** First delivery to Singapore Airlines.
- * **October 25, 2007:** Entered service.
- * **2012 & 2014:** Production peaked at 30 aircraft per year.
- * **2021:** Manufacturing ended.

Challenges:

- * Initial production was delayed by two years due to electrical wiring issues.
- * Development costs almost doubled, failing to be recouped by the end of production.

Operational Status

Current Fleet:

- * As of April 2024, there were 189 aircraft in service with 10 operators worldwide.
- * Fifteen airlines operated the A380, with five having fully retired the aircraft from their fleets.

Performance:

- * As of December 2021, the global A380 fleet had completed over 800,000 flights, totaling 7.3 million block hours.
- * The A380 has maintained a safety record with no fatalities and no hull losses.

Design and Engineering

Manufacturing and Transportation:

- * Major structural sections were built in France, Germany, Spain, and the United Kingdom.

- * Specialized road and water transportation methods were developed due to the size of the components.
- * Components were transported to the Jean-Luc Lagardère Plant assembly hall in Toulouse, France.

Testing and Certification:

- * Five A380s were built for testing and demonstration purposes.
- * The first A380 (F-WWOW) was unveiled on January 18, 2005, and first flew on April 27, 2005.
- * Achieved maximum design speed of Mach 0.96 during testing.
- * Underwent high-altitude and cold-weather testing in various locations.
- * Received EASA and FAA approval to carry up to 853 passengers on March 29, 2006.

Production Issues

- * Initial production faced delays due to wiring complexity, design and production concurrency, high customization, and configuration management failures.
- * CATIA version incompatibilities between different Airbus facilities added to the issues.

Variants

A380-800

- * The primary passenger model.

- * Original configuration: 555 passengers in three classes.
- * Maximum capacity: 853 passengers in a single-class configuration.
- * Design range: 8,500 nmi (15,700 km).

A380F (Freighter)

- * Planned cargo variant; development was suspended.
- * Capacity: 150 t (330,000 lb) of cargo.
- * Range: 5,600 nmi (10,400 km).

Proposed Variants

- * **A380-900 Stretch:** A stretched version seating approximately 656 passengers (or up to 960 in all-economy).
- * **A380neo:** A proposed new engine option, aimed at improving fuel efficiency.
- * **A380plus:** An enhanced variant with lower costs per seat and improved fuel economy.

Image

![[Airbus A380 cross section]](https://upload.wikimedia.org/wikipedia/commons/thumb/3/3b/Airbus_A380_cross_section.svg/250px-Airbus_A380_cross_section.svg.png)

Image description: An illustration of the cross section of an Airbus A380, showing the two decks and passenger seating.

Performance and Specifications

Engines

- * Rolls-Royce Trent 900 (A380-841/-842)
- * Engine Alliance GP7000 (A380-861)

Noise Reduction

- * Modern engines and aerodynamic improvements result in lower noise levels.
- * Complies with noise limits at major airports like London Heathrow.

Advanced Materials

- * Composite materials comprise over 20% of the airframe.
- * Hybrid fiber metal laminate material GLARE used in the upper fuselage.
- * Newer weldable aluminum alloys used in the airframe.

Avionics and Systems

- * Integrated modular avionics (IMA) architecture.
- * Fly-by-wire flight controls.
- * Advanced glass cockpit with liquid crystal displays.
- * Power-by-wire flight control actuators.
- * 350 bar (5,000 psi) hydraulic system.

- * Variable-frequency electrical generators.

Cabin Features

- * 550 square meters (5,920 sq ft) of usable floor space.
- * Features to reduce traveler fatigue, such as a quieter interior and higher pressurization.
- * Seating options ranging from first-class suites to 11-abreast economy.
- * LED lighting system.

Airport Compatibility

- * Designed to operate safely on Group V runways and taxiways.
- * Requires service vehicles with lifts capable of reaching the upper deck.

Image

![[Airbus A380 Cockpit]](https://upload.wikimedia.org/wikipedia/commons/thumb/a/a4/Airbus_A380_cockpit.jpg/250px-Airbus_A380_cockpit.jpg)

Image description: A view of the Airbus A380 cockpit, showcasing its advanced avionics and display systems.

Airline Operations and Economics

Initial Operators

- * Singapore Airlines: First A380 entered service on October 25, 2007.
- * Emirates: Commenced service between Dubai and New York in August 2008.
- * Qantas: Flights between Melbourne and Los Angeles in October 2008.

Operational Issues

- * Wing cracks discovered during repairs led to inspections and modifications.
- * Door leaks resulted in emergency landings and costly replacements.

Production Cuts and Financial Challenges

- * Production cuts announced due to reduced deliveries.
- * Analysts questioned whether the A380 project would ever break even.

Secondary Market

- * Several airlines expressed interest in selling or leasing their aircraft.
- * Malaysia Airlines decided to refurbish its A380s for religious pilgrimage flights.
- * Hi Fly leased used aircraft for markets needing high capacity.

Order and Delivery Details

Customers

- * Emirates is the largest customer, with a total of 123 A380s ordered.
- * Fourteen customers ordered and took delivery of the A380.

Cancellations

- * Skymark Airlines, Qantas, Amedeo, and Virgin Atlantic cancelled their orders.
- * Emirates reduced its order in 2019, leading to the end of A380 production.

Final Production

- * Airbus ended A380 production in 2021, with the last aircraft delivered to Emirates.

Cumulative Orders and Deliveries

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|---|---|

|Total Orders| 251|

|Aircraft in service (May 2025)| 189 with 12 operators|

Incidents and Accidents

- * The A380 has never been involved in a hull-loss accident as of January 2025.
- * Notable incidents include uncontained engine failures.

Related Information

Related Development

Aircraft of comparable role, configuration, and era

Related lists

This documentation provides a comprehensive overview of the Airbus A380, covering its development, features, operational history, and more.